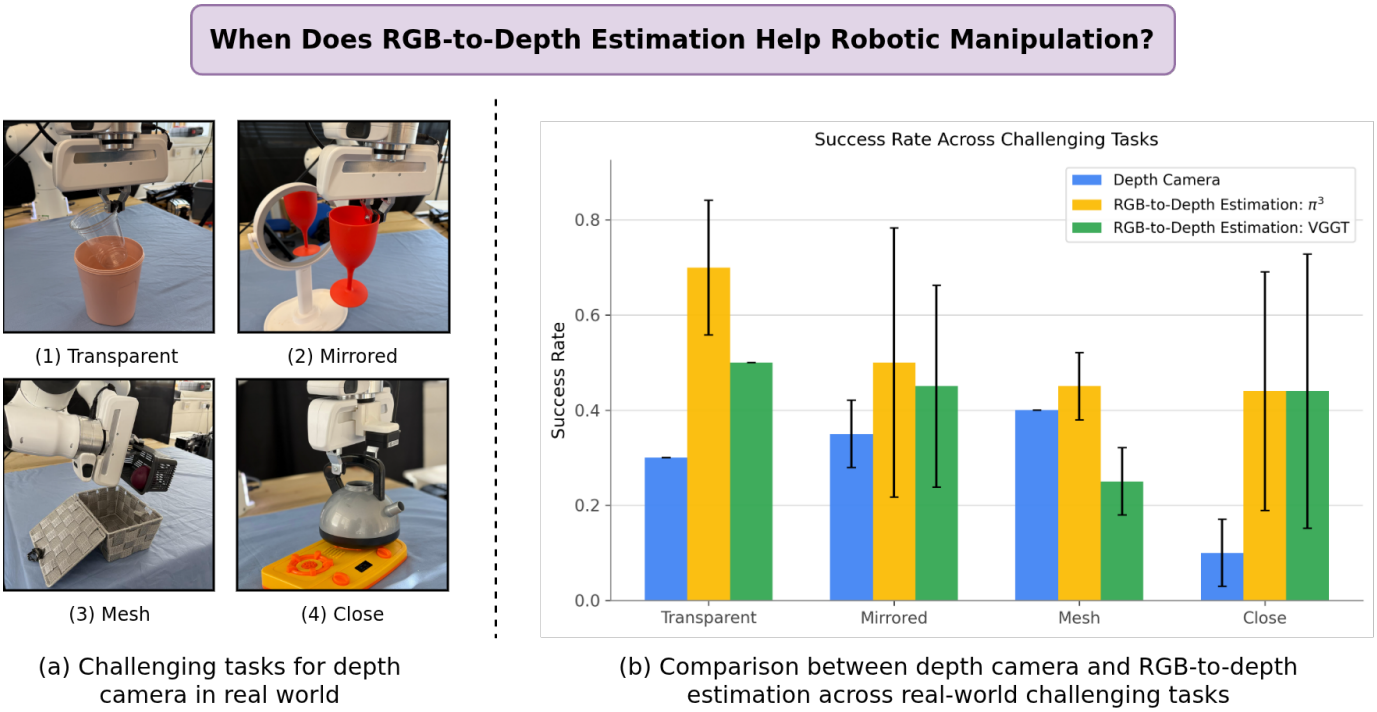
Depth Camera or RGB-to-Depth?
A Study of RGB-to-Depth Estimation for Robot Policy Learning

Yifei Ren[1], Pietro Vitiello[1], and Edward Johns[1]



(a) Challenging tasks for depth camera in real world

(b) Comparison between depth camera and RGB-to-depth estimation across real-world challenging tasks

Fig. 1: In robot policy learning pipelines, depth images are commonly used to construct point clouds, enabling explicit 3D reconstruction and scene understanding. However, depth sensors often fail in challenging settings such as transparent, mirrored, mesh, and close-range scenes (left). We found that learned depth estimation from RGB images achieves comparable or higher success rates across these tasks (right), indicating that explicit depth sensing is not always optimal and that depth estimation can serve as an effective alternative in challenging conditions.

***Abstract*—Point cloud–based robot policy learning has emerged as a powerful approach for robotic manipulation by enabling policies to directly exploit 3D geometric structure. In such pipelines, depth image acquisition plays a critical role in point cloud construction. However, raw depth from depth cameras is often noisy, sensitive to transparent, reflective, and repetitive surfaces, and limited by a minimum sensing distance. Recent advances in learned depth estimation offer a promising alternative: modern monocular, stereo, and multi-view approaches have achieved impressive performance in vision benchmarks. However, robotics still lacks a systematic understanding of how these advances translate to end-to-end policy learning and execution. In this work, based on our introduced Depth-Estimated Imitation Learning (DEIL) general framework, we present the first systematic study of how modern depth estimation impacts point cloud–based imitation learning across two paradigms: In-Context Imitation Learning (ICIL) and Behaviour Cloning (BC). Within each paradigm, we keep the point cloud reconstruction procedure, policy backbone, and evaluation protocol fixed across depth sources. Through 71,800 simulated and 160 real-world rollouts, several depth estimators achieve reasonable downstream performance in simulation, although a consistent gap to GT depth remains. In real-world ICIL, the best-performing multi-view estimator is competitive with the tested RGB-D cameras on normal tasks and achieves higher success when depth sensing is unreliable. Our failure mode analysis further shows that reconstructed geometric quality generally correlates with policy success, with flying points being particularly harmful to BC. These findings highlight the need for improved geometric awareness and stability before learned depth estimation can be broadly deployed in robot manipulation. Videos and Appendix can be found on our webpage: https://flycole.github.io/deil.github.io/.**

***Index Terms*—Performance Evaluation and Benchmarking, Perception for Grasping and Manipulation, Imitation Learning.**

Manuscript received: April 29, 2026; Revised August 4, 2026; Accepted August 28, 2026.

This paper was recommended for publication by Editor Lucia Pallottino upon evaluation of the Associate Editor and Reviewers' comments. This work was supported by the Royal Academy of Engineering under the Research Fellowship Scheme.

[1]Yifei Ren, Pietro Vitiello and Edward Johns are with the Robot Learning Lab at Imperial College London. {yifei.ren20, pietro.vitiello17, e.johns}@imperial.ac.uk

Digital Object Identifier (DOI): see top of this page.

I. Introduction

Modern robotic manipulation increasingly depends on the quality of visual observations used to represent the scene. While many state-of-the-art behaviour cloning (BC) [1], [2], [3], [4] and vision-language-action (VLA) [5], [6], [7], [8], [9] methods primarily operate on RGB observations, a growing line of work [10], [11], [12], [13], [14] incorporates 3D geometric representations, such as point clouds, to better capture spatial structure and object relationships that are difficult to infer from raw RGB inputs alone. As a result, the choice of how to obtain these 3D observations becomes a central question for downstream robotic manipulation.

In point cloud–based policy learning, 3D observations are traditionally obtained using depth sensors, which directly provide per-pixel depth measurements and enable straightforward construction of point cloud representations. Such sensors have been widely adopted in robotics due to their simplicity and tight integration with perception pipelines. However, depth cameras also exhibit well-known limitations: they are often noisy, require a minimum sensing distance to the target object (below which depth measurements become invalid), and struggle with challenging visual properties such as reflective surfaces, repetitive mesh-like structures, and transparent objects. These issues can lead to incomplete or corrupted point clouds, directly affecting downstream policy learning and execution.

Learned monocular, pairwise, and multi-view depth estimators [15], [16], [17], [18], [19], [20] offer an alternative by inferring geometry from RGB observations, often using large-scale pretraining or multi-view constraints. However, existing benchmarks primarily measure pixel-level accuracy or static reconstruction quality, leaving their downstream effects on robot policy learning underexplored and providing limited guidance for estimator selection.

To address this gap, we systematically evaluate how representative RGB-to-depth estimators affect point cloud-based imitation learning. Our study compares learned depth with ground-truth (GT) depth in simulation and with the tested RGB-D cameras in the real world, while also examining failure modes and computational costs. We use Depth-Estimated Imitation Learning (DEIL) as a controlled evaluation pipeline across two representative paradigms: **In-Context Imitation Learning (ICIL)**, which conditions policies on the contexts of the demonstrations at inference time, and **Behaviour Cloning (BC)**, which learns from demonstration data. We select Instant Policy (IP) [13] for ICIL and 3D Diffusion Policy (DP3) [12] for BC because both natively consume point clouds. Within each paradigm, we keep the point cloud reconstruction procedure, policy backbone, and evaluation protocol fixed.

In simulation, the best-performing learned estimators achieve reasonable downstream performance under both paradigms, although a consistent gap to GT depth remains. In real-world ICIL, strong multi-view estimators remain competitive with the tested RGB-D cameras on normal tasks and improve success when depth sensing is unreliable, including transparent, reflective, repetitive-texture, and close-range settings. Our failure analysis on BC associates lower success rates with poorer reconstructed geometry. Among the evaluated metrics, point cloud accuracy error, which captures spurious or flying points, shows the strongest correlation with success rate. Timing results further reveal substantial estimator-dependent latency, with depth estimation and point-cloud reconstruction accounting for most of the runtime.

II. Related Work

Behaviour Cloning. Imitation learning has been studied extensively for acquiring robot skills from expert demonstrations. Classical behaviour cloning (BC) [1], [21] learns task-specific policies through supervised action prediction but typically requires hundreds of demonstrations per task. To improve data efficiency and generalisation, subsequent work has explored richer model architectures and learning objectives. ACT [4] predicts action chunks using a CVAE-based Transformer, while RT-1 [5] formulates discretised action prediction as token generation. Diffusion Policy [3] instead learns action distributions through diffusion, effectively capturing multimodal behaviours in high-dimensional action spaces. Other studies [2], [22], [23], [24], [25] further extend imitation learning to one-shot and zero-shot settings. Despite their strong empirical performance, these methods primarily rely on 2D RGB observations and thus only implicitly encode 3D geometric structure through deep visual features. However, manipulation often requires explicit spatial reasoning about object poses, contact geometry, and free space. This has motivated policies that operate on explicit 3D representations, including point clouds, voxels, and multi-view reconstructions. 3D Diffusion Policy (DP3) [12] extends diffusion-based action modelling to point cloud observations using a dedicated 3D encoder, while PerAct [11] and RVT [10] use voxelised scene representations and 3D tokens for spatial reasoning. Together, these approaches demonstrate the benefits of explicit 3D geometry for robot policy learning.

In-Context Imitation Learning. While the above methods achieve strong performance, they typically rely on task-specific training or large-scale pretraining and cannot rapidly adapt to new tasks from only a few demonstrations. In-Context Imitation Learning (ICIL) instead conditions policies directly on demonstrations at test time, enabling new behaviours without gradient-based fine-tuning. Early ICIL methods explored structured representations for effective context conditioning. KAT [26] encodes demonstrations as sequences of keypoint-based action tokens, allowing a single policy to generalise across tasks through contextual reasoning. Subsequently, [27] directly tokenises observations and formulates robot control as next-token prediction with a Transformer. Instant Policy [13] introduces a novel GNN-based in-context architecture together with a pseudo-demonstration generation scheme for scalable training, enabling direct action inference for novel tasks. Collectively, these methods highlight the potential of combining 3D perception with in-context learning for rapid adaptation. However, the effect of the depth-acquisition and reconstruction pipeline used to construct their point cloud inputs remains underexplored.

RGB-to-Depth Estimation. Depth estimation has undergone rapid progress in recent years, driven by advances in

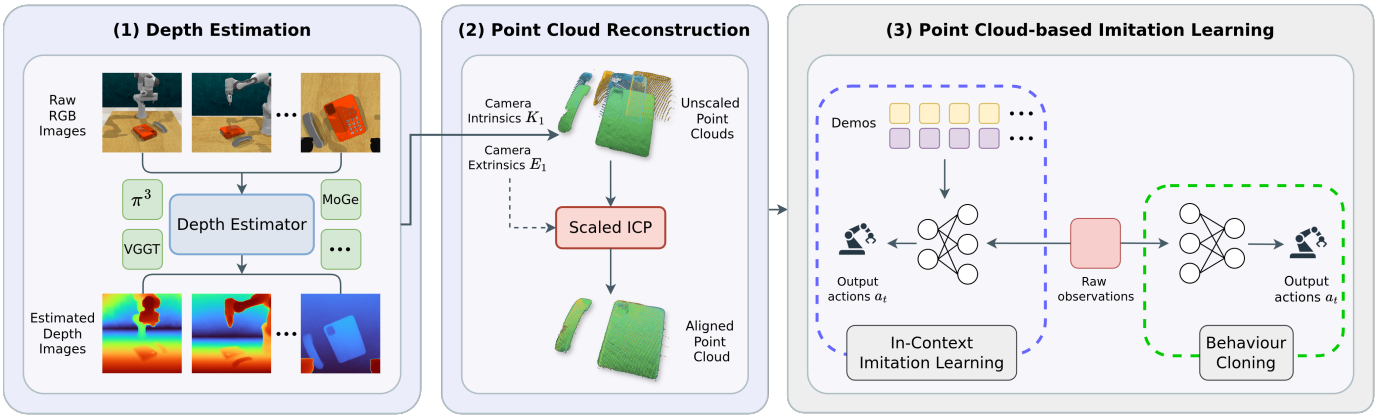


Fig. 2: The pipeline of Depth-Estimated Imitation Learning. The framework consists of three stages. First, in the depth estimation stage, raw RGB images are passed through a depth estimator to produce corresponding depth maps. Second, using the camera intrinsics and extrinsics, unscaled point clouds are reconstructed from each camera view, and a Scaled ICP method is applied to align and merge them into a unified representation. Third, point cloud–based behaviour cloning or in-context imitation learning methods are trained on the reconstructed 3D representations generated by the previous modules.

large-scale pretraining and geometric learning. Existing approaches can be broadly categorised by their input assumptions and geometric constraints. *Monocular depth estimation*, such as MoGe [17], Apple Depth Pro [28], Depth Anything [29], [30], and Metric3D [31], infer depth from a single RGB image and therefore allow flexible deployment without multi-view inputs. However, these approaches often struggle to recover metric depth and may lack cross-view consistency when independently applied to multiple cameras. *Stereo depth estimation*, such as Dust3r [18] and MASt3R [19], jointly reconstruct geometry from image pairs, improving depth accuracy and relative pose estimation [32] but requiring pairwise processing and alignment for more than two views. In contrast, *multi-view depth estimation*, such as π^3 [15] and VGGT [16], jointly process multiple images to produce more coherent scene geometry in cluttered or occluded environments. MapAnything [20] can additionally condition on known camera geometry when available, to support metric scale recovery and scene reconstruction. Recently, generative 3D pipelines [33], [34], [35], [36] aim to synthesise consistent 3D representations from sparse visual inputs, enabling indirect depth estimation. However, these methods are still relatively immature for robotic manipulation, so we leave their evaluation to future work.

III. DEIL: Depth-Estimated Imitation Learning

A. Overview

To systematically study the role of depth estimation in imitation learning, particularly under challenging perceptual conditions, we propose Depth-Estimated Imitation Learning (DEIL) as a modular evaluation framework. As illustrated in Fig. 2, DEIL consists of three components: depth estimation, point-cloud reconstruction, and imitation learning. This modular design allows us to plug in different depth estimation methods while holding the learning framework fixed, enabling controlled evaluation of their impact on policy performance.

B. Depth Estimation

We incorporate representative state-of-the-art depth estimators covering four categories with different input assumptions and geometric constraints. For each camera C_i, we obtain its intrinsic parameters K_i, extrinsic parameters E_i, and RGB observation I_i. Depth estimation is formulated as

$$\{\tilde{Z}_i\} = \Phi(\{I_i, E_i\}), \tag{1}$$

where $\tilde{Z}_i$ is the estimated depth map and Φ denotes the depth estimator. Camera extrinsics are provided only to methods that explicitly require them, such as MapAnything.

C. Point Cloud Reconstruction

Each estimated depth map is unprojected into the world coordinate frame:

$$\tilde{P}_i^{\text{unscaled}} = \pi^{-1}(\tilde{Z}_i, K_i, E_i). \tag{2}$$

However, many depth estimators recover only relative geometry, resulting in scale inconsistencies and misalignment across views. We therefore apply scaled ICP to align the reconstructed point clouds while keeping the calibrated camera poses fixed. For stereo-based methods such as MASt3R, more than two views are handled using loop alignment across view pairs. All methods otherwise share the same point cloud reconstruction pipeline for fair comparison. For implementation details, please refer to our website.

D. Imitation Learning

We evaluate depth estimation under two paradigms: In-Context Imitation Learning (ICIL) and Behaviour Cloning (BC). Because our objective is to evaluate depth estimators through downstream manipulation, both policies must natively consume point clouds. We therefore select Instant Policy (IP) [13] and 3D Diffusion Policy (DP3) [12], which are well-established and competitive methods for ICIL and BC, respectively. For both paradigms, the reconstructed point cloud is the sole geometric input, while the remaining policy pipeline is kept fixed.

IV. EXPERIMENTS

A. Experimental Setup

In this work, we evaluate depth estimation methods across two imitation learning paradigms: In-Context Imitation Learning (ICIL) and Behaviour Cloning (BC). Experiments are conducted in both simulation and the real world. Simulation covers both paradigms, enabling efficient comparisons across tasks and hyperparameters. Real-world evaluation is limited to ICIL because collecting approximately 100 demonstrations per task for BC is prohibitively expensive. For both paradigms, we consider six depth variants: (a) **Depth Camera**: Raw RGB-D camera measurements in the real world and GT depth as an upper bound in simulation. (b) **MoGe**: A fast monocular estimator without multi-view constraints, but lacking natural cross-view consistency. (c) **MASt3R**: A pairwise geometry estimator that uses correspondences to improve consistency across views. (d) $\boldsymbol{\pi^3}$: A multi-view model with a permutation-equivariant architecture that predicts affine-invariant camera poses and scale-invariant local pointmaps. (e) **VGGT**: A feed-forward multi-view model that jointly predicts camera parameters, depth maps, pointmaps, and tracks. (f) **MapAnything**: Depth estimation with known camera extrinsics. All experiments are conducted using an NVIDIA RTX 4090 GPU for data collection, policy training, and inference.

B. Low-Budget Hyperparameter Screening

Depth estimation quality depends on several implementation choices that act as hyperparameters, which directly affect the quality of the reconstructed point clouds and downstream policy performance. We consider three of them:

- Camera configuration: front; front + side; front + side + wrist.
- Input resolution: 128, 256, and 512.
- Image masking: whether irrelevant image regions are masked.

However, exhaustively evaluating all estimators and configurations would require 216,000 rollouts, which is computationally expensive and exceeds our evaluation budget.

We therefore conduct low-budget screening in simulation under the lightweight ICIL protocol. Each configuration is evaluated on 10 RLBench tasks [37]. For each configuration-task pair, we use five demonstrations as separate contexts and perform two rollouts per demonstration, yielding 10 rollouts. We average the success rates across tasks to obtain the overall screening score. Complete results are available on our website.

The results show that π^3 and VGGT consistently outperform the other estimators across configurations, approaching depth camera performance in several settings, whereas MoGe, MASt3R, and MapAnything are less accurate and stable, particularly with single-view or low-resolution inputs. Masking has estimator-dependent effects, providing modest improvements in some settings but negligible or negative effects in others. Separately, resolution is not the primary performance bottleneck. Increasing the resolution from 128 to 512 yields diminishing performance gains for most methods once multi-view coverage is available.

Based on the screening results, we adopt the following configuration-selection strategy for the main experiments: (i) we evaluate multiple shortlisted configurations of π^3 and VGGT under the full protocol and report all results; and (ii) for each remaining estimator, we use its best screening configuration and keep it fixed across tasks. We note that the selected configurations are specific to this task set and may not be optimal for unseen tasks. This strategy allows us to focus computational resources on the most promising methods while maintaining a fair and practical comparison across baselines.

C. Simulation Experiments

We conduct the main simulation experiments on the same 10 manipulation tasks from RLBench [37] as before, evaluating on both ICIL and BC paradigm. All evaluations use newly sampled demonstrations and rollout seeds independent of those used during screening.

1) In-Context Imitation Learning: For the main experiments under ICIL, we collect 20 demonstrations per task. Conditioning each demonstration as a single context, we run Instant Policy [13] for 10 rollouts based on point clouds constructed through each depth variation. We report the average success rate across demonstrations, rollouts and tasks as the final performance metric in Table I. Note that for the depth camera baseline, we also report the all-camera setting (5 cameras), which is the same as in the original paper for fair comparison.

Depth Estimator	Camera Setting	Res.	Mask	SR	SR Std.
	Depth Sensor				
Depth Camera	front	128	None	0.760	0.238
Depth Camera	front	256	None	0.746	0.273
Depth Camera	front	512	None	0.756	0.257
Depth Camera	front / side	128	None	0.933	0.052
Depth Camera	front / side	256	None	0.930	0.050
Depth Camera	front / side	512	None	0.925	0.053
Depth Camera	front / side / wrist	128	None	0.878	0.139
Depth Camera	front / side / wrist	256	None	0.875	0.164
Depth Camera	front / side / wrist	512	None	0.875	0.169
Depth Camera	all	128	None	0.952	0.045
Depth Camera	all	256	None	0.938	0.063
Depth Camera	all	512	None	0.934	0.074
	Depth Estimation				
π^3	front / side	256	True	0.701	0.234
π^3	front / side	512	True	0.696	0.231
π^3	front / side	256	False	0.741	0.192
π^3	front / side	512	False	0.722	0.189
π^3	front / side / wrist	256	False	0.612	0.271
π^3	front / side / wrist	512	False	0.631	0.243
VGGT	front / side	256	True	0.725	0.235
VGGT	front / side	512	True	0.722	0.250
VGGT	front / side	256	False	0.734	0.182
VGGT	front / side	512	False	0.631	0.180
VGGT	front / side / wrist	256	False	0.660	0.240
VGGT	front / side / wrist	512	False	0.571	0.241
MoGe	front / side	256	False	0.496	0.266
MASt3R	front / side	256	False	0.113	0.091
MapAnything	front / side	256	False	0.481	0.246
MapAnything*	front / side	256	False	0.505	0.191

Note: MapAnything* stands for MapAnything without known camera extrinsics.

TABLE I: In-context imitation learning (Instant Policy) performance under different depth estimation pipelines. We report success rate (SR) and standard deviation across rollouts. For π^3 and VGGT, we highlight the best-performing configuration in the fine-grained search.

As shown, it highlights the strong dependence of ICIL performance on the quality and configuration of the depth

Tasks	**Depth**	π^3	**VGGT**	**MoGe**	**MASt3R**	**MA**	**Noisy$_{10mm}$**	**Noisy$_{20mm}$**	**Noisy$_{30mm}$**	**Noisy$_{40mm}$**
Phone On Base	1.00	0.96	**0.98**	0.96	0.72	0.26	1.00	1.00	1.00	1.00
Open Box	0.98	**0.74**	0.40	0.50	0.56	0.12	0.98	0.98	1.00	0.88
Close Box	1.00	**0.96**	0.64	0.72	0.54	**0.96**	0.98	0.98	0.98	0.94
Plate Out	0.70	**0.30**	0.24	0.26	0.06	0.20	0.66	0.46	0.42	0.34
Basketball	1.00	**0.72**	0.62	0.46	0.3	0.36	0.94	0.94	0.94	0.90
Close Microwave	0.78	**0.54**	0.34	**0.54**	0.36	0.14	0.76	0.80	0.74	0.74
Toilet Seat Down	0.90	**0.90**	0.80	**0.90**	0.64	0.46	0.86	0.84	0.84	0.86
Toilet Seat Up	0.90	0.66	0.20	**0.78**	0.56	0.16	0.84	0.62	0.46	0.46
Toilet Roll Off	0.40	0.28	0.08	**0.28**	0.48	0.04	0.56	0.36	0.36	0.24
Meat Off Grill	1.00	0.90	0.96	0.94	0.94	**0.98**	1.00	0.98	1.00	1.00
Average	0.87	**0.70**	0.53	0.63	0.52	0.37	0.86	0.80	0.77	0.74
Standard Deviation	0.19	0.25	0.32	0.26	0.24	0.34	0.15	0.24	0.26	0.28

TABLE II: Behaviour cloning (DP3) performance under different depth estimation and corrupted-depth variations. We report the success rate across all tasks and rollouts. Noisy$_*$ is corrupted-depth variations in different corruption levels. Depth is short for GT Depth and MA is short for MapAnything.

estimation methods. Obviously, using depth from the depth sensor consistently yields the highest success rates, and moving from a single front view to multi-view settings leads to substantial gains. However, increasing resolution from 128 to 512 provides a slight performance drop, indicating that geometric coverage is more critical than image resolution within ICIL.

Depth estimation narrows but does not close the gap compared to the ground truth depth. Among all depth estimators, π^3 [15] and VGGT [16] achieve the strongest performance, consistently outperforming other depth estimation methods. This result underscores the importance of multi-view geometric consistency for downstream control, as methods that explicitly leverage multi-view cues produce point clouds that better support spatial alignment in ICIL. Nevertheless, a noticeable gap remains compared to the ground truth depth, indicating that current depth estimators still struggle to provide the geometric fidelity required for high-precision manipulation.

2) Behaviour Cloning: In the BC paradigm, we use a *front / side* camera setup at 256 resolution, and apply no image masking for all depth variations. For each task, we collect 100 demonstrations and generate depth images using each depth variant. To assess BC sensitivity to depth noise, we also add four corrupted-depth baselines by applying Gaussian noise with standard deviations of 10, 20, 30, and 40mm to ground-truth (GT) depth. Together with the six original depth variants, we train ten types of DP3 policies. For each policy, we evaluate up to seven checkpoints at 20-epoch intervals, each using 50 rollouts with the same set of seeds across checkpoints and depth variants, and select the best-performing checkpoint. Average success rates across tasks are reported in Table II.

As shown, GT depth achieves an average success rate of 87%, serving as an upper-bound reference. The evaluated depth estimators exhibit substantially different performance across tasks. Among them, π^3 achieves the highest average success rate of 70%, although with a slightly higher standard deviation than GT depth, demonstrating the strong overall performance of multi-view depth estimation for downstream manipulation. MoGe achieves the second-best performance, with a 27.6% relative reduction compared with GT depth, indicating the potential of monocular depth estimation as well. The remaining methods achieve average success rates ranging from 37% to 54% and exhibit larger variations across tasks. Further improvements in stability and robustness are therefore required before these estimators can reliably replace hardware depth sensors.

Additionally, the corrupted-depth baselines further verify that DP3 is sensitive to depth and reconstruction noise. When the noise standard deviation increases to 40mm, the average success rate decreases to 73.6%, approaching the performance of several depth estimators. This sensitivity is also strongly task-dependent. For relatively easy tasks without substantial occlusion, such as *Phone on Base* and *Close Box*, even the highest corruption level has little effect on success. In contrast, for more difficult tasks or those involving substantial occlusion, such as *Plate Out*, *Toilet Seat Up*, and *Toilet Roll Off*, increasing the noise standard deviation from only 10 to 20mm can already cause a considerable performance degradation. For task details and reconstructed point clouds during rollouts, please check on our website.

3) Failure Mode Analysis: To quantify how depth estimation affects policy performance, we compare each estimated point cloud with the GT reference at every visited state during DP3 rollouts. Metrics are computed per frame, averaged per rollout, and analysed against success rates. We consider three complementary metrics: **Accuracy,** the mean estimated-to-reference nearest-neighbour distance in millimeters. It measures if estimated points lie near the true surfaces; larger values indicate inaccurate or flying points, spurious geometry. **Completeness,** the mean reference-to-predicted nearest-neighbour distance in millimeters. It measures how well the estimated point cloud covers the true geometry; larger values indicate missing surfaces, holes, or lost thin structures. **F-score,** the harmonic mean of precision and recall at each distance threshold. It summarises overall reconstruction quality by requiring the point cloud to contain both accurate predictions and complete geometric coverage.

Fig. 3 visualises all of the metric values in a geometric profile heatmap. It consists of two panels: depth estimation and corrupted-depth baselines. Both panels summarise the task-balanced performance of each depth variation. Rollout-

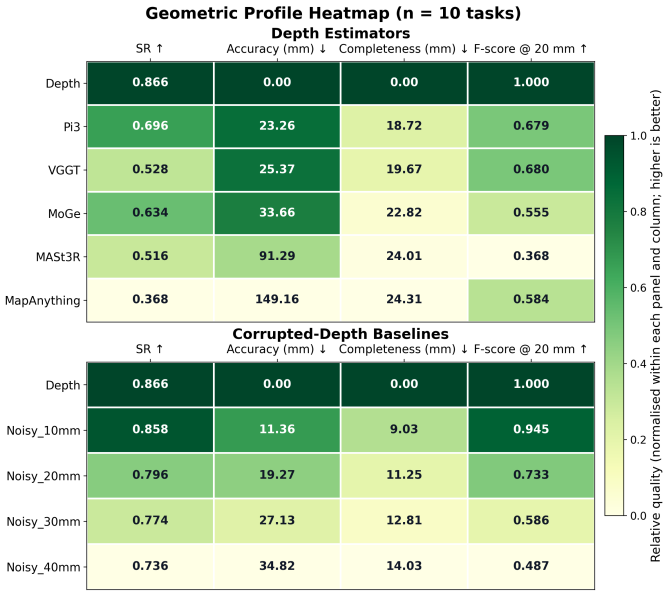


Fig. 3: Geometric profile heatmap across ten tasks. The upper panel compares depth estimators, while the lower panel evaluates corrupted-depth baselines at increasing noise levels. Each cell displays the raw mean value for success rate (SR), accuracy, completeness, and F-score at 20 mm, while colours are normalised independently for each metric, with higher intensity consistently indicating better performance. GT Depth is included in this panel as a reference.

level metrics are first averaged within each task and then averaged equally across tasks, preventing tasks with more steps from dominating. As shown, the corrupted-depth baselines exhibit a consistent relationship between geometric quality and policy performance: as the noise level increases from 10 to 40mm, SR decreases from 0.858 to 0.736, accompanied by higher accuracy and completeness errors and a lower F-score. Among the depth estimators, π^3 achieves the lowest accuracy and completeness errors, corresponding to the highest SR of 0.696. Accuracy is particularly informative for severe outliers: MapAnything has an accuracy error of 149.16mm and the lowest SR of 0.368, despite a moderate F-score of 0.584. Completeness also reflects performance degradation: MoGe has a completeness error of 22.82mm and an SR of 0.634, compared with 14.03mm and 0.736 for $\text{Noisy}_{40\text{mm}}$. However, no metric independently determines success. For example, MoGe achieves a higher SR than VGGT (0.634 vs. 0.529) despite having worse accuracy, completeness, and F-score. Therefore, these metrics reflect general SR trends, but their relationship with success must be interpreted jointly and conditioned on the error structure and task characteristics.

We further compute the Pearson and Spearman correlations between each metric and success rate. Accuracy exhibits the strongest Pearson correlation and is therefore visualised in Fig. 4, where accuracy closely tracks SR, indicating that accuracy error is strongly associated with DP3 performance.

4) System Latency Analysis: Table III reports the average per-step control latency across four stages: depth estimation (Est.), point-cloud reconstruction (Recon.), policy-input preprocessing (Preproc.), and policy inference (Inference). MoGe

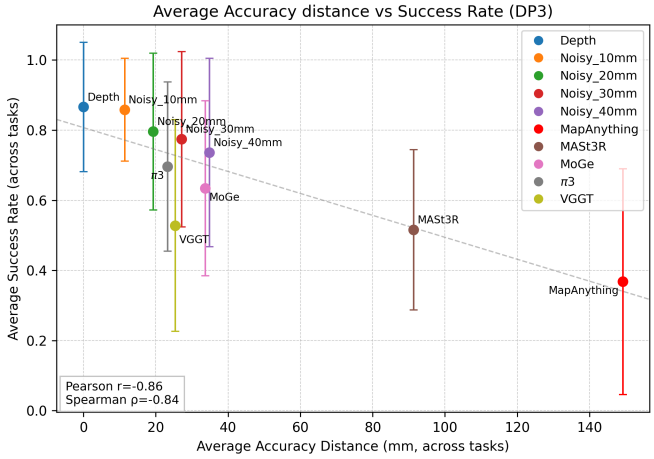


Fig. 4: Influence of Accuracy on task success rate. Each coloured point represents a different depth estimator or corrupted-depth baseline, with error bars indicating the standard deviation across tasks. The dashed line shows the fit to the data points in the figure.

achieves the lowest overall latency, followed closely by π^3, whereas MASt3R is substantially slower because of its loop-alignment stage. Reconstruction, preprocessing, and policy-inference costs remain similar across estimators, making depth estimation the primary source of latency variation. The MoGe and π^3 based pipelines operate at approximately 1Hz, while the others are slower. Further optimisation of the depth models and system implementation could move the pipeline closer to real-time operation.

	Est.	Recon.	Preproc.	Inference	Total
$\boldsymbol{\pi^3}$	273	458	145	47	923
VGGT	892	468	146	56	1562
MoGe	261	428	143	40	872
MASt3R	2286	422	141	40	2889
MA	789	486	145	40	1460

TABLE III: Average per-step control latency breakdown in milliseconds. MA denotes MapAnything.

D. Real World Experiments

We evaluate DEIL in the real world on a set of 16 manipulation tasks, illustrated in Fig. 5. For the RGB-D sensors, we adopt a common robot learning setup: two Intel RealSense D455 cameras for the *front* and *side* views, which provide reliable depth at their operating distances, and one D435 camera for the *wrist* view, whose shorter minimum sensing distance is better suited to close-range manipulation. We further select specific camera configuration according to the requirements of each task. To assess performance under diverse sensing conditions, we categorise these tasks into two levels: **normal** and **challenging**.

The **challenging** category covers four common RGB-D failure modes: **transparent**, **mirrored**, **mesh**, and **close**. The first three represent transparency, reflection, and repetitive-pattern challenges, respectively. We include two tasks per type using a

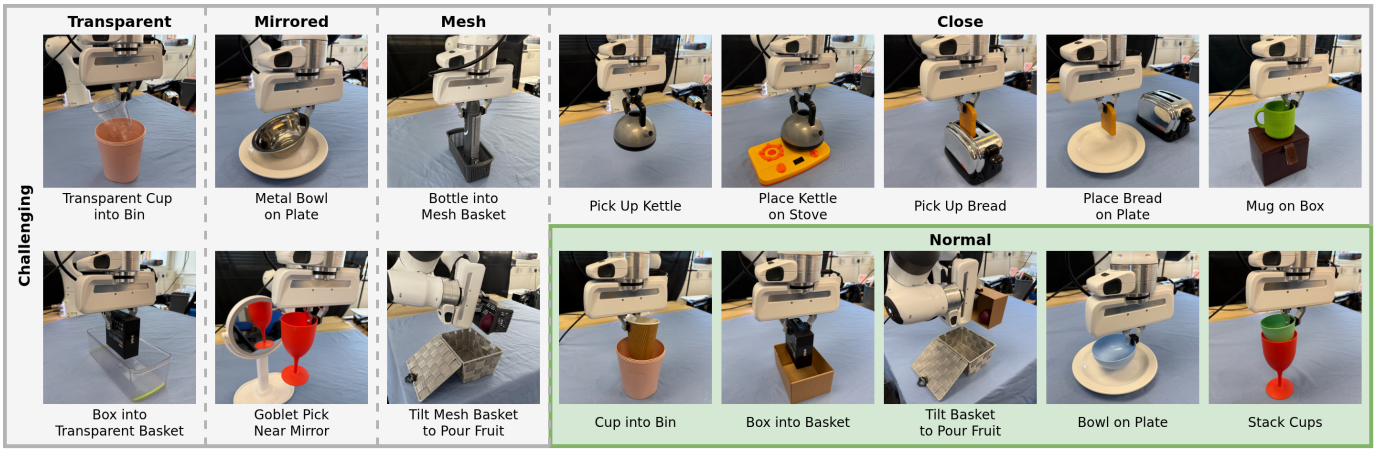


Fig. 5: Illustration of the 16 real-world tasks used in our evaluation. We evaluate DEIL across two categories: normal and challenging. The normal category comprises five tasks (green block). The challenging category includes four types of variations: Transparent (2 tasks), Mirrored (2 tasks), Mesh (2 tasks), and Close (5 tasks).

front / side camera setup consistent with simulation. The **close** type contains five tasks using a *front / wrist* setup. As the wrist camera approaches the object during manipulation, the object-to-camera distance may fall below the sensor's minimum operating range, resulting in invalid depth measurements. The **normal** category contains five tasks without extreme visual properties as a baseline and uses the same *front / side* setup.

Due to the high cost of large-scale real-world demonstration collection, we restrict real-world experiments to ICIL. For each task, we evaluate Instant Policy using point clouds constructed from three depth sources: the two best-performing estimators in simulation, π^3 and VGGT, and raw camera depth as a baseline. Following the original Instant Policy protocol, we use two demonstrations as context and conduct 10 rollouts per task, each limited to 30 execution steps. Figure 6 reports the average success rates.

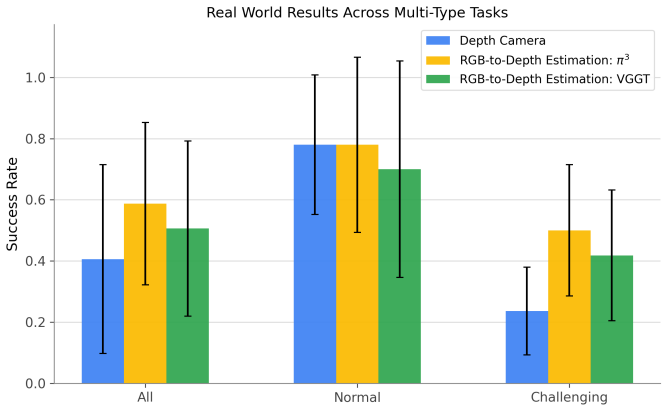


Fig. 6: Real-world results across different categories. We compare Depth Camera, π^3, and VGGT using average success rate, with standard deviation indicated by error bars.

Overall, π^3 achieves the strongest and most consistent performance. It matches the Depth Camera baseline on normal tasks while maintaining higher success rates under challenging conditions, where the tested depth cameras degrade substantially. VGGT provides moderate improvements over Depth Camera but is less consistent than π^3.

The task-variant results in Fig. 1 (right) further show that π^3 outperforms both baselines across all evaluated challenging conditions, with the largest gains for transparent objects and close-distance settings. These results demonstrate its greater robustness to the visual ambiguities and sensing failures examined in our real-world evaluation.

V. Conclusion

Using DEIL as a controlled evaluation pipeline, we present the first systematic study of how RGB-to-depth estimation affects point cloud-based imitation learning. In simulation, the depth estimators could achieve reasonable downstream performance under both ICIL and BC, although a consistent gap to GT depth remains. Our failure analysis associates lower success rates with poorer reconstructed geometry. Among the evaluated metrics, point cloud accuracy error, which captures spurious or flying points, shows the strongest correlation with success rate. The timing results further reveal substantial estimator-dependent latency, with depth estimation and point cloud reconstruction accounting for most of the runtime. In real-world ICIL, strong multi-view estimators match RGB-D sensing on normal tasks and improve performance when the tested active-stereo cameras are unreliable, including transparent, reflective, repetitive-texture, and close-range settings. These findings suggest that learned depth estimation can complement or replace hardware depth sensing in challenging settings. However, future methods should prioritise geometrically robust and computationally efficient reconstruction.

VI. Limitations

Our study has several methodological limitations. We evaluate two representative imitation-learning paradigms with fixed policy backbones, so the findings may not generalise to other architectures, embodiments, or control settings. The experiments cover a limited range of manipulation tasks and camera configurations, and the real-world evaluation is restricted to ICIL on a single robotic setup. Although we use two RealSense models (D435 and D455), both rely on the same

active-stereo sensing principle. Other RGB-D technologies, such as structured-light and time-of-flight sensing, may exhibit different failure modes; thus, our conclusions about hardware depth sensing apply only to the tested RealSense setup. Our point-cloud failure analysis provides a more hardware-agnostic basis for comparison, but broader validation across sensing technologies is needed. Moreover, the proposed metrics summarise global point cloud quality and may not fully characterise specific reconstruction failure modes. Future work should examine additional policies, robots, camera arrangements, depth sensing technologies, and real-world BC settings, while developing more accurate metrics that better capture these failure modes.

ACKNOWLEDGMENT

The authors would like to thank Cheng Qian, Yi Zhao, Vitalis Vosylius, Yilong Wang, Ruomeng Fan, and Ivan Kapelyukh for their valuable discussions related to this work.

REFERENCES

[1] T. Zhang, Z. McCarthy, O. Jow, D. Lee, X. Chen, K. Goldberg, and P. Abbeel, "Deep imitation learning for complex manipulation tasks from virtual reality teleoperation," in *2018 IEEE international conference on robotics and automation (ICRA)*. Ieee, 2018.

[2] E. Jang, A. Irpan, M. Khansari, D. Kappler, F. Ebert, C. Lynch, S. Levine, and C. Finn, "Bc-z: Zero-shot task generalization with robotic imitation learning," in *Conference on Robot Learning*. PMLR, 2022, pp. 991–1002.

[3] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," *The International Journal of Robotics Research*, p. 02783649241273668, 2023.

[4] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware," in *Proceedings of Robotics: Science and Systems*, Daegu, Republic of Korea, July 2023.

[5] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, J. Dabis, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog, J. Hsu *et al.*, "Rt-1: Robotics transformer for real-world control at scale," *arXiv preprint arXiv:2212.06817*, 2022.

[6] B. Zitkovich, T. Yu, S. Xu, P. Xu, T. Xiao, F. Xia, J. Wu, P. Wohlhart, S. Welker, A. Wahid *et al.*, "Rt-2: Vision-language-action models transfer web knowledge to robotic control," in *Conference on Robot Learning*. PMLR, 2023, pp. 2165–2183.

[7] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi *et al.*, "Openvla: An open-source vision-language-action model," *arXiv preprint arXiv:2406.09246*, 2024.

[8] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter *et al.*, "π_0: A vision-language-action flow model for general robot control," *arXiv preprint arXiv:2410.24164*, 2024.

[9] P. Intelligence, K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai *et al.*, "$\pi_{0.5}$: a vision-language-action model with open-world generalization," *arXiv preprint arXiv:2504.16054*, 2025.

[10] A. Goyal, J. Xu, Y. Guo, V. Blukis, Y.-W. Chao, and D. Fox, "Rvt: Robotic view transformer for 3d object manipulation," in *Conference on Robot Learning*. PMLR, 2023, pp. 694–710.

[11] M. Shridhar, L. Manuelli, and D. Fox, "Perceiver-actor: A multi-task transformer for robotic manipulation," in *Conference on Robot Learning*. PMLR, 2023, pp. 785–799.

[12] Y. Ze, G. Zhang, K. Zhang, C. Hu, M. Wang, and H. Xu, "3D Diffusion Policy: Generalizable Visuomotor Policy Learning via Simple 3D Representations," in *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024.

[13] V. Vosylius and E. Johns, "Instant policy: In-context imitation learning via graph diffusion," *arXiv preprint arXiv:2411.12633*, 2024.

[14] K. Dreczkowski, P. Vitiello, V. Vosylius, and E. Johns, "Learning a thousand tasks in a day," *Science Robotics*, vol. 10, no. 108, 2025.

[15] Y. Wang, J. Zhou, H. Zhu, W. Chang, Y. Zhou, Z. Li, J. Chen, J. Pang, C. Shen, and T. He, "π^3: Permutation-equivariant visual geometry learning," *arXiv preprint arXiv:2507.13347*, 2025.

[16] J. Wang, M. Chen, N. Karaev, A. Vedaldi, C. Rupprecht, and D. Novotny, "Vggt: Visual geometry grounded transformer," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 5294–5306.

[17] R. Wang, S. Xu, C. Dai, J. Xiang, Y. Deng, X. Tong, and J. Yang, "Moge: Unlocking accurate monocular geometry estimation for open-domain images with optimal training supervision," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 5261–5271.

[18] S. Wang, V. Leroy, Y. Cabon, B. Chidlovskii, and J. Revaud, "Dust3r: Geometric 3d vision made easy," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 20 697–20 709.

[19] V. Leroy, Y. Cabon, and J. Revaud, "Grounding image matching in 3d with mast3r," in *European Conference on Computer Vision*. Springer, 2024, pp. 71–91.

[20] N. Keetha, N. Müller, J. Schönberger, L. Porzi, Y. Zhang, T. Fischer, A. Knapitsch, D. Zauss, E. Weber, N. Antunes *et al.*, "Mapanything: Universal feed-forward metric 3d reconstruction; map-anything. github. io," in *2026 International Conference on 3D Vision (3DV)*. IEEE, 2026, pp. 499–509.

[21] S. Young, D. Gandhi, S. Tulsiani, A. Gupta, P. Abbeel, and L. Pinto, "Visual imitation made easy," in *Conference on Robot learning*. PMLR, 2021, pp. 1992–2005.

[22] E. Johns, "Coarse-to-fine imitation learning: Robot manipulation from a single demonstration," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

[23] I. Kapelyukh, Y. Ren, I. Alzugaray, and E. Johns, "Dream2Real: Zero-shot 3D object rearrangement with vision-language models," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

[24] G. Papagiannis and E. Johns, "Miles: Making imitation learning easy with self-supervision," in *Proceedings of the Conference on Robot Learning (CoRL)*, 2024.

[25] Y. Wang and E. Johns, "One-shot dual-arm imitation learning," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 5660–5668.

[26] N. D. Palo and E. Johns, "Keypoint Action Tokens Enable In-Context Imitation Learning in Robotics," in *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024.

[27] L. Fu, H. Huang, G. Datta, L. Y. Chen, W. C.-H. Panitch, F. Liu, H. Li, and K. Goldberg, "In-context imitation learning via next-token prediction," *arXiv preprint arXiv:2408.15980*, 2024.

[28] A. Bochkovskiy, A. Delaunoy, H. Germain, M. Santos, Y. Zhou, S. Richter, and V. Koltun, "Depth pro: Sharp monocular metric depth in less than a second," in *International Conference on Learning Representations*, Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu, Eds., vol. 2025, 2025, pp. 75 602–75 637.

[29] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, and H. Zhao, "Depth anything: Unleashing the power of large-scale unlabeled data," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 10 371–10 381.

[30] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, "Depth anything v2," *Advances in Neural Information Processing Systems*, vol. 37, pp. 21 875–21 911, 2024.

[31] W. Yin, C. Zhang, H. Chen, Z. Cai, G. Yu, K. Wang, X. Chen, and C. Shen, "Metric3d: Towards zero-shot metric 3d prediction from a single image," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 9043–9053.

[32] Y. Wang, C. Qian, R. Fan, and E. Johns, "Observer actor: Active vision imitation learning with sparse view gaussian splatting," *arXiv preprint arXiv:2511.18140*, 2025.

[33] X. Kong, S. Liu, X. Lyu, M. Taher, X. Qi, and A. J. Davison, "Eschernet: A generative model for scalable view synthesis," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 9503–9513.

[34] J. Zhou, H. Gao, V. Voleti, A. Vasishta, C.-H. Yao, M. Boss, P. Torr, C. Rupprecht, and V. Jampani, "Stable virtual camera: Generative view synthesis with diffusion models," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2025, pp. 12 405–12 414.

[35] J. Xu, W. Cheng, Y. Gao, X. Wang, S. Gao, and Y. Shan, "Instantmesh: Efficient 3d mesh generation from a single image with sparse-view large reconstruction models," *arXiv preprint arXiv:2404.07191*, 2024.

[36] S. Liu, K. W. Ng, W. Jang, J. Guo, J. Han, H. Liu, Y. Douratsos, J. Perez, Z. Zhou, K. C. Phung, T. Xiang, and J.-M. Perez-Rua, "Scaling sequence-to-sequence generative neural rendering," in *International Conference on Learning Representations*, C. Vondrick, B. Hariharan, C. Raffel, L. Pinto, D. Yang, and A. Faust, Eds., vol. 2026, 2026, pp. 65 298–65 323.

[37] S. James, Z. Ma, D. R. Arrojo, and A. J. Davison, "Rlbench: The robot learning benchmark & learning environment," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 3019–3026, 2020.